

ALI VAYANI

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Education

University of Texas at Austin

Bachelor of Science in Computer Science

May 2028

GPA: 3.6

Experience

Aramco

May 2025 – Aug 2025

Software Engineer Intern

Houston, Texas

- Spearheaded the design and implementation of an **LLM-powered knowledge retrieval system** supporting **100+ employees** across **5+ departments**, streamlining access to internal documentation
- Developed a robust backend using **Python, Flask, LangChain**, and **ChromaDB**, embedding and indexing **500+ internal documents** as dense vectors for high-performance and low-latency semantic search
- Engineered a distributed retrieval engine from scratch, integrating **semantic sectioning**, **contextual chunk headers**, and **segment-level post-processing** to outperform vanilla RAG approaches on long-form, multi-step queries by **35%**
- Designed a **multi-agent LLM architecture** for risk reporting: **Coordinated specialized agents** (research, analysis, drafting) using **LangGraph**, projected to reducing report drafting time by **50%**
- Partnered with infrastructure and security teams to ensure **compliance with internal governance**, while designing for future **on-premise deployment and scale**

CodeWiz

Apr 2024 – Aug 2024

Software Engineer Intern

Keller, Texas

- Developed and deployed **Node.js automation tools** to streamline schedule creation, boosting team efficiency by **15%**
- Built an internal dashboard for real-time schedule visibility and admin control, improving oversight and decision-making speed
- Used **Puppeteer** to scrape and consolidate scheduling data from various sources, increasing coverage by **30%**
- Automated employee-to-class assignments based on dynamic data inputs, leading to smarter **resource allocation** and a **25% drop in scheduling errors**

Organizations & Extracurricular

MyMosque

May 2025 – Present

Tech Lead

Austin, Texas

- Leading a team of **5+ student developers** to build a mosque management platform with **mobile app (Expo/React Native)** and **admin web portal (Next.js/AWS Lambda)**, using **Supabase** for backend services while focusing on mentorship
- Launched beta with **100+ weekly users**, implementing **CI/CD pipelines**, real-time analytics, push notifications, and multi-mosque support to ensure scalability and a production-ready experience

Freetail Hackers

Jan 2025 – Present

Software Engineer

Austin, Texas

- Built and maintained core features for **Rodeo**, a hackathon management platform supporting **1500+ applicants** and **200+ project submissions**, using **Svelte, Prisma**, and **tRPC** to streamline check-ins, judging, and submissions
- Designed and shipped **Apple** and **Google Wallet** integrations used by **1150+ participants**, enabling instant ticketing and digital hacker IDs — contributing to a smoother day-of-event flow and reducing check-in bottlenecks by **30%+**

Longhorn Developers

Aug 2024 – Present

Software Engineer

Austin, Texas

- Contributed to **UT Registration Plus**, a Chrome extension with **60,000+ users**, helping ship features like a **real-time visual indicator** for course availability across **5,000+ classes**
- Built the “**Microwave Map**” feature for **UT Dining**, a mobile app with **5,000+ daily users**, using **React Native (Expo)** to display all campus microwaves and guide users to the nearest one based on their real-time location

Projects

The Heat Ordering Platform, Fullstack | NextJS, React, Firebase, TailwindCSS

- Built a responsive, user-friendly interface with **React**, enabling staff to browse menus, manage carts, and place detailed orders — resulting in **500+ orders** and over **\$2.5K in revenue** within a single semester
- Built a secure **admin dashboard** with **Firebase auth** and real-time tools for managing orders and updating live menus, streamlining daily operations

Transformer-based Song Lyrics Generator | PyTorch, Pandas

- Built a **GPT-style transformer model** from scratch in **PyTorch** to generate original song lyrics, implementing multi-head attention, positional encoding, and token-level generation
- Trained the model on a **custom dataset of 10,000+ lyrics**, using gradient clipping and learning rate scheduling to improve coherence and **reduce loss by 30%** over training

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, C++, C#, HTML/CSS

Frameworks & Libraries: React, Next.js, Flask, React Native, PyTorch, Pandas, NumPy, TailwindCSS, LangChain, ChromaDB

Tools & Platforms: Git, GitHub, Docker, VS Code, Android Studio, Postman, GCP, AWS, Figma